SPLITTING OF SYMPLECTIC BRAID-GROUP MONODROMY FOR TYPE A MILNOR FIBRES IN EVERY COMPLEX DIMENSION AT LEAST THREE

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Allowed standard prerequisites (editorial).

- Autoequivalence groups and connected stability spaces of type A Calabi-Yau categories. Ikeda Theorems7.15,7.16; Qiu-Woolf Theorems B; BS Theorem9.9, all cited precisely in author proof
- Cluster categories and exchange-graph quotient. Baur-Marsh, BRT, King-QiuYu-cluster Theorem8.6
- Koszul duality and Rabinowitz/cluster equivalence for Dynkin plumbings. AbSm-plumbing, Kalck-Yang, BJK Corollary1.4 and Lekili-Ueda as explicitly cited

Source reference. Original section 4: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 665–665. A footnote points to the later discussion for an analogue of the already explicitly cited AbSm-plumbing/Kalck-Yang Koszul duality fact; the Milnor case is established via those named references and the complete local splitting chain is included.

Source reference. Original section 7: algebra-geometry/sources/sigma-2026-080/native/sigma26-080.tex, source lines 1307–1307. Forward pointer in introductory categorical scope only, unrelated to the type A splitting proof.

1. Flavours of Fukaya category

Let (X, ω) be a connected symplectic manifold with $2c_1(X) = 0$. We will fix a Maslov cover of the Lagrangian Grassmannian, meaning that there is a well-defined group $\mathrm{Symp}^{\mathrm{gr}}(X)$ of graded symplectomorphisms, and a subgroup $\mathbb{Z} \leqslant \mathrm{Symp}^{\mathrm{gr}}(X)$ generated by the shift (i.e., the fibrewise deck transformation). Recall from [74, Lemma 2.4] the following.

Lemma 1.1. *There is an exact sequence*

$$1 \to \mathbb{Z} \to \mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X) \to H^1(X; \mathbb{R}).$$

The final map is not a homomorphism but its kernel is a subgroup.

Lemma 1.2. *Suppose that X is non-compact. The forgetful map $\mathrm{Symp}^{\mathrm{gr}}(X) \to \mathrm{Symp}(X)$ is split over the subgroup $\mathrm{Symp}_{\mathrm{ct}}(X)$.*

Proof. Take the unique gradings for symplectomorphisms which act trivially on the fibres of the Maslov cover near infinity. □

Let (X, ω) be a Liouville manifold, with contact type boundary, and assume $2c_1(X) = 0$. We will be interested in all of

(1) the compact Fukaya category $\mathcal{F}(X)$, with objects compact graded exact spin Lagrangians;

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(2) the wrapped Fukaya category $\mathcal{W}(X)$, with objects graded exact spin Lagrangians which are cones over Legendrian submanifolds near infinity;
(3) the Rabinowitz Fukaya category $\mathcal{R}(X) := \mathcal{W}(X)/\mathcal{W}(X)^{\mathrm{prop}}$, which is the Verdier quotient of the (Yoneda image of the) wrapped category by its subcategory of proper modules.

All three categories are defined over $\mathbb{Z}$, and $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a full and faithful subcategory. We say that 'Koszul duality' holds for the compact and wrapped categories if $\mathcal{F}(X) \subset \mathcal{W}(X)$ is a Koszul dualising subcategory in the sense of [26, Definition 2.12]. This often holds for Weinstein manifolds obtained from plumbings of simply connected pieces. Concretely, the wrapped category of a Weinstein manifold admits a finite set of generators $\{T_i \mid 1 \leqslant i \leqslant N\}$ and one seeks a finite set of compact branes $\{L_i \mid 1 \leqslant i \leqslant N\}$ such that the total endomorphism algebras $\mathcal{Q} = \bigoplus_{i,j} HF^*(L_i, L_j)$ and $\mathcal{A} = \bigoplus_{i,j} HW^*(T_i, T_j)$, which are augmented over the semisimple ring $\Bbbk = \bigoplus_{i=1}^N \mathbb{K}_i$ generated by the units of the objects, are A_∞-Koszul dual, meaning

$$\mathcal{Q} \simeq \mathrm{Ext}_{\mathcal{A}}(\Bbbk, \Bbbk), \qquad \mathcal{A} \simeq \mathrm{Ext}_{\mathcal{Q}}(\Bbbk, \Bbbk).$$

In this setting, there is an equivalence [26, Corollary 1.4]

$$\mathcal{R}(X) \simeq \mathcal{W}(X)/\mathcal{F}(X).$$

This is the formulation we will encounter in practise, but the general algebraic construction of $\mathcal{R}(X)$ from $\mathcal{W}(X)$ in particular shows that autoequivalences of the latter act on the former. (In Section 7, it will be important to work over $\mathbb{C}$, to incorporate $\mathbb{C}^*$-valued local systems on Lagrangians.)

Any autoequivalence of $\mathcal{W}(X)$ arising from a symplectomorphism of X will preserve the compact category $\mathcal{F}(X)$, and hence induce an equivalence of $\mathcal{R}(X)$. For background on $\mathcal{F}(X)$, see [76], and for background on $\mathcal{W}(X)$, see [1]; for background on derived localisation in general, see [22], and in the context of $\mathcal{R}(X)$ specifically, see [26].

The Grothendieck group $K(\mathcal{F}(X))$ is the free abelian group on isomorphism classes of objects, modulo relations coming from exact triangles. For later use, we define the 'numerical Grothendieck group'

$$K_{\mathrm{num}}(\mathcal{F}(X)) \tag{1.1}$$

to be the quotient of $K(\mathcal{F}(X))$ by the subgroup which is the radical of the Euler pairing, i.e., that generated by compact exact Lagrangians $L \in \mathrm{Ob}\,\mathcal{F}(X)$ for which $\chi(HF^*(L, \bullet)) = 0$ for every $\bullet \in \mathcal{W}(X)$.

We will be interested in actions of symplectomorphism groups on Fukaya categories. Especially for actions on the wrapped category, one needs some control of maps at infinity to ensure that (perhaps after isotopy) they preserve the class of Lagrangians which are conical[1] at infinity. We say a symplectomorphism ϕ of a Liouville manifold $(X, \mathrm{d}\theta)$ is *exact* if $\phi^*\theta = \theta + \mathrm{d}f$ for some $f\colon X \to \mathbb{R}$, and *strongly exact* if one can take f to have compact support. Both notions have obvious graded analogues. It is classical that strongly exact maps act on the wrapped category. Write $\mathrm{Symp}_{\mathrm{ex}}(X)$ for the group of exact symplectomorphisms.

Lemma 1.3. *Assume that the Liouville manifold X has finite type, so can be written as the completion of a Liouville domain $\mathring{X}$. Then*

(1) *There is a homomorphism $\pi_0 \mathrm{Symp}_{\mathrm{ct}}(X) \to \pi_0 \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$;*
(2) *The group $\pi_0 \mathrm{Symp}(\mathring{X})$ acts on both $\mathcal{F}(X)$ and $\mathcal{W}(X)$.*

[1]Or more generally geometrically bounded for a dissipative almost complex structure in the sense of Groman [29].

Proof. For the first statement, conjugation by a suitably large Liouville flow will bring the support into $\mathring{X}$. We claim we can then deform to obtain an exact representative of the map. Let $\omega = \mathrm{d}\lambda$ be the Liouville form on X, and suppose we are given $\psi \in \mathrm{Symp}_{\mathrm{ct}}(X)$. We have $\psi^*\lambda - \lambda = \theta$, where θ is a closed one-form with compact support. Let V be the vector field which is symplectically dual to $-\theta$. This has compact support, and so is integrable for all time; let σ_t denote its flow. Then applying Cartan's formula, we get that $\psi \circ \sigma_1$ is an exact symplectomorphism. (Note that the proof of [11, Lemma 1.1] specialises to this argument in this case.)

This construction can be performed in families, which shows that the inclusion

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \hookrightarrow \mathrm{Symp}_{\mathrm{ct}}(X)$$

is a homotopy equivalence, so induces an isomorphism on π_0. On the other hand, there is then a forgetful map

$$\mathrm{Symp}_{\mathrm{ct,ex}}(X) \to \mathrm{Symp}_{\mathrm{ex}}(\mathring{X})$$

which induces a homomorphism on π_0. Combining these two gives the first statement.

The second statement is exactly [45, Corollary 2.4] (cf. [46], which in particular involves deforming a map of the Liouville domain to be exact, using a version of the argument given in the previous paragraphs). □

Lemma 1.4. *In the setting of Lemma* 1.3*, assume moreover that* $2c_1(\mathring{X}) = 0$*. Then there are compatible representations*

$$\begin{array}{ccc} \pi_0\,\mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X) & \longrightarrow & \mathrm{Auteq}(\mathcal{F}(X)) \\ \downarrow & & \uparrow \\ \pi_0\,\mathrm{Symp}^{\mathrm{gr}}(\mathring{X}) & \longrightarrow & \mathrm{Auteq}(\mathcal{W}(X)). \end{array}$$

The top line is classical; the bottom line is [45, Corollary 2.9] (see also [46]).

Although the Rabinowitz category is not a priori determined by the contact boundary of X, one still has the following lemma.

Lemma 1.5. *The group* $\pi_0\,\mathrm{Symp}^{\mathrm{gr}}_{\mathrm{ct}}(X)$ *acts trivially on the Rabinowitz category.*

Proof. Let L, K be Lagrangians which are cones over Legendrian submanifolds L_∞ and K_∞. The morphism group $RFC^*(L,K)$ in $\mathcal{R}(X)$ between L and K is the chain-level mapping cone of a natural continuation map $CW_*(L,K) \to CW^*(L,K)$ coming from a monotone homotopy of Hamiltonians from 'infinite negative wrapping to infinite positive wrapping'. This factors as a composition

$$CW_*(L,K) \xrightarrow[a]{} CF^*(L,\phi_{-\varepsilon}K) \xrightarrow[c]{} CF^*(L,\phi_{+\varepsilon}K) \xrightarrow[b]{} CW^*(L,K),$$

where a, b are projection respectively inclusion maps of summands into the relevant direct systems and c is the natural continuation from small negative to small positive wrapping. Viewing $RFC^*(L,K) = \mathrm{Cone}(b \circ c \circ a)$, there are exact triangles of mapping cones

$$\mathrm{Cone}(c \circ a) \to RFC^*(L,K) \to \mathrm{Cone}(b)$$

and

$$\mathrm{Cone}(a) \to \mathrm{Cone}(c \circ a) \to \mathrm{Cone}(c).$$

The continuation map c induces the Poincaré duality isomorphism in Lagrangian Floer theory

$$HF^*(L,\phi_{-\varepsilon}K) \to HF^*(L,\phi_{+\varepsilon}(K))$$

(which recovers the isomorphism $H_*(L,\partial_\infty L) \to H^{n-*}(L)$ when $L = K$), so has acyclic cone. On the other hand, $\mathrm{Cone}(b) \simeq CW(L,K)^+$ defines the Lagrangian

Floer analogue of 'positive symplectic cohomology' [70]; it can be computed by an action truncated version of the wrapped Floer complex which removes action zero chords (i.e., interior intersection points), yielding a complex only built from Reeb chords at infinity between the Legendrian boundaries. Similarly for $\mathrm{Cone}(a)$. Putting these ingredients together, one obtains a cochain complex quasi-isomorphic to that underlying $RFC^*(L,K)$ and which is generated only by Reeb chords at the boundary. Any compactly supported map then acts trivially on the underlying chain group.

Since L and K were arbitrary, compactly supported maps act trivially on the whole A_∞-category. □

To conclude this section, we briefly recall the categories associated to quivers with potential; a more extensive discussion can be found in [83].

A quiver Q is a directed graph, and a potential W on Q is a formal sum of cycles in the edges. To such (Q,W) one can associate a 3-dimensional[2] Calabi–Yau category $\mathcal{C}(Q,W)$ via the 'Ginzburg construction' [28], which defines a non-positively graded dg-algebra $\mathcal{J}(Q,W)$ with $H^0(\mathcal{J}(Q,W)) = J(Q,W)$ the Jacobi algebra (i.e., the path algebra modulo cyclic derivatives of the potential). We always assume that W has no quadratic terms; then the category $\mathcal{C}$ has a spherical object for each vertex of Q, and W encodes the non-trivial products in a cyclic A_∞-structure on the total endomorphism algebra of those objects (this is the Koszul dual of the Ginzburg algebra). By a 'symplectic geometric model' for $\mathcal{C}(Q,W)$ we mean a configuration of Lagrangian spheres $\{L_v\}_{v\in \mathrm{Vert}(Q)} \subset (X,\omega)$ such that the number of edges between v and v' gives the rank of $HF^*(L_v, L_{v'})$ and the endomorphism algebra $\bigoplus_{v,v'} HF^*(L_v, L_{v'})$ is A_∞-quasi-isomorphic to that determined by the potential W. Such models are especially useful when the $\{L_v\}$ (split-)generate the compact category $\mathcal{F}(X,\omega)$. One can also seek algebro-geometric models for $\mathcal{C}(Q,W)$, i.e., embeddings $\mathcal{C}(Q,W) \hookrightarrow \mathcal{D}(Y)$ into the derived category of some quasi-projective Y (ideally with image some naturally characterised subcategory), or indeed algebraic models in terms of realisations in derived categories of modules over a non-commutative algebra. All of these will play a role in the sequel.

2. Type $\boldsymbol{A}$ Milnor fibres

We consider the $A_k^{(n)}$-Milnor fibre of complex dimension n, that is the affine hypersurface

$$\left\{x_1^2 + \cdots + x_n^2 + p_{k+1}(z) = 0\right\} \subset \left(\mathbb{C}^{n+1}, \omega_{\mathrm{st}}\right) \tag{2.1}$$

with p_{k+1} a degree $k+1$ polynomial with distinct zeroes. This is symplectically a plumbing of k copies of T^*S^n, associated to the A_k-quiver:

Figure 1. The A_k-quiver.

It comes in a family over the configuration space $\mathrm{Conf}_{k+1}(\mathbb{C})$ parametrizing polynomials of degree $k+1$ with distinct zeroes, which yields a representation

$$\rho\colon\ \mathrm{Br}_{k+1} \to \pi_0 \,\mathrm{Symp}_{\mathrm{ct}}\left(A_k^{(n)}\right)$$

from the classical braid group on $k+1$ strings. Combining the work of many authors one has the following assertion.

Proposition 2.1 ([49, Corollary 1.4])**.** *The representation ρ is injective for every k, n.*

[2]When the arrows admit suitable gradings, one can associate a CY_n-category for any n.

Proposition 2.2 ([23, Theorem 1.4] and [89, Theorem 1.2])**.** *When $n = 2$, ρ is an isomorphism.*

Proposition 2.3 ([76, Corollary 17.17] and [18, Theorem A])**.** *For some pairs k, n the monodromy ρ is not surjective.*

The first result is based on direct computations of Floer cohomology; the second relies on special features of holomorphic curves in four dimensions; the third result uses 'reframings' of twists by exotic diffeomorphisms of spheres (for instance, it applies when $n = 8$ and $k \in \{1, 2\}$). This last result makes the following interesting statement.

Theorem 2.4. *For $n \geqslant 3$ and arbitrary k, the map ρ splits.*

We will assume $n \geqslant 3$ throughout the rest of this section. We first establish the slightly simpler proposition.

Proposition 2.5. *There is a map $p\colon \pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big) \to \mathrm{Br}_{k+1}/Z(\mathrm{Br}_{k+1})$ for which the composite $p \circ \rho$ is the natural projection.*

Proof. Since $\pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big)$ maps canonically to $\pi_0 \operatorname{Symp}^{\mathrm{gr}}\big(A_k^{(n)}\big)$, it acts on the space of stability conditions on the compact Fukaya category $\mathcal{F}$. This has a distinguished component in the stability space coming from tilts of chambers associated to the standard heart. Qiu and Woolf [67, Theorem B] prove that in fact the space of stability conditions is connected, so the whole of $\mathrm{Auteq}(\mathcal{F})$ acts on this unique component. The autoequivalences preserving the given component have been computed in [12, Theorem 9.9] when $n = 3$ and in general in [38, Theorem 7.16] and [67, Theorem B]. The complete answer is as follows. We can identify $\mathcal{F}$ with the category of modules over the endomorphism algebra of the A_k-chain of Lagrangian 'core' spheres. Ikeda proves that

$$\mathrm{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big) = (\mathrm{Br}_{k+1} \times \mathbb{Z})/\mathbb{Z}, \tag{2.2}$$

where the $\mathbb{Z}$-factor in the numerator is generated by the shift, and the embedding of the denominator sends the generator to $(\Delta, d_{k,n})$ where Δ generates the centre and

$$d_{k,n} = (k+1)(n-2) + 2. \tag{2.3}$$

There are thus exact sequences

$$1 \to \mathrm{Br}_{k+1} \to \mathrm{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big) \to \mathbb{Z}/d_{k,n}\mathbb{Z} \to 1;$$

and

$$1 \to \mathbb{Z} \to \mathrm{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big) \to \mathrm{Br}_{k+1}/Z(\mathrm{Br}_{k+1}) \to 1$$

and the latter admits a map from the sequence

$$1 \to \mathbb{Z} \to \pi_0 \operatorname{Symp}^{\mathrm{gr}}_{\mathrm{ct}}\big(A_k^{(n)}\big) \to \pi_0 \operatorname{Symp}_{\mathrm{ct}}(A_k^{(n)}) \to 1,$$

where we are using the fact that the forgetful map $\pi_0 \operatorname{Symp}^{\mathrm{gr}}_{\mathrm{ct}}\big(A_k^{(n)}\big) \to \pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big)$ splits by Lemma 1.2. The existence of a map $\pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big) \to \mathrm{Br}_{k+1}/Z(\mathrm{Br}_{k+1})$ follows.

The map ρ takes the i-th standard generator of the braid group to the Dehn twist in the i-th Lagrangian sphere in the A_k chain which is the compact core of $A_k^{(n)}$. Under Ikeda's isomorphism (2.2), the braid group factor is generated by the corresponding spherical twists. That implies that $p \circ \rho$ is the natural projection. □

Remark 2.6. Since the centre of the braid group acts by a shift on $\mathcal{F}(X)$, it acts trivially on any simplicial complex constructed from the set of (ungraded) Lagrangian spheres in X. One can detect the centre of the braid group through its action on the ungraded category underlying the partially wrapped Fukaya category studied in [76, Section 20], which adjoins a single Lefschetz thimble to the A_k-chain of spheres and wraps 'infinitesimally'. However, the space of stability conditions on this partially wrapped Fukaya category has not been computed.

Via the splitting of the map from graded to ungraded symplectomorphisms over the compactly supported subgroup of Lemma 1.2, we have a representation

$$\rho\colon\ \pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big) \to \operatorname{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big)$$

and to prove Theorem 2.4 it remains to show that the composite $\chi\circ\rho\colon \pi_0 \operatorname{Symp}_{\mathrm{ct}}\big(A_k^{(n)}\big) \longrightarrow \mathbb{Z}/d_{k,n}\mathbb{Z}$ vanishes. This will follow directly from a geometric picture of the target.

Recall the cluster category $\mathcal{C}_{k,n}$ of type (k,n) is by definition the orbit category of the path category of the underlying quiver of $A_k^{(n)}$, given by dividing out by the composition $\tau^{-1}[n-2]$ of the given power of the shift and the inverse of the Auslander–Reiten translation τ, see, e.g., [8]. It is naturally a CY_{n-1}-category.

Recall the integer $d_{k,n}$ introduced in (2.3). Let (S,M) be the marked bordered surface which is the disc S with a collection of $d_{k,n}$ boundary marked points, equivalently of a regular $d_{k,n}$-gon. We think of the group $\mathbb{Z}/d_{k,n}\mathbb{Z}$ as the oriented mapping class group $\Gamma(S,M)$ of (S,M).

Lemma 2.7. *There is a natural homomorphism*

$$\operatorname{Auteq}(\mathcal{C}_{k,n}) \longrightarrow \Gamma(S,M) = \mathbb{Z}/d_{k,n}\mathbb{Z}.$$

Proof. This is a variation on the proof of [12, Theorem 9.9], see also [38, Theorem 7.15], and is exactly the result [13, Theorem 4.7] in the special case when the cluster category is CY_2. The paper [8] gives a combinatorial description of the cluster category. Its indecomposable objects are given by the n-diagonals of a $d_{k,n}$-gon (i.e., diagonals which decompose the polygon into an n-gon and a remainder) and the cluster tilting objects biject with n-angulations of the $d_{k,n}$-gon (maximal sets of disjoint-in-the-interior n-diagonals; the diagonals may intersect at the boundary, and any such maximal set has k members). See [8, Proposition 5.4] and [40, Theorem 4.2]. (In the special case, where the cluster category is CY_2, one studies actual triangulations of the polygon.) An illustration with $n=4$, $k=3$ is given in Figure 2. There is a (directed) 'flip' orientation on n-angulations: a single n-diagonal is a diameter in a unique $(2n-2)$-gon, and one can rotate the ends of the given diagonal one place clockwise in this subpolygon to obtain a new n-angulation, cf. the labelled edge δ contained in the shaded $(2n-2)$-gon in Figure 2 and the forward flip along δ on the right.

The cluster exchange graph of $\mathcal{C}_{k,n}$ (with vertices being simple hearts and edges arising from tilts) then has a purely combinatorial description in terms of directed flips of n-angulations, cf. [8, Theorem 5.6] or [38, Theorem 6.3].

The orientation-preserving mapping class group of the marked disc acts (with finite stabilisers) on n-angulations and hence on the cluster exchange graph.

The A_k-quiver has an associated CY_n-category, via the Ginzburg construction, which also has a heart exchange graph. It is a non-trivial theorem of Buan, Reiten and Thomas [15] that one can associate to a heart in the Ginzburg category a 'silting collection' in the quotient cluster category, and that this defines a map from the heart exchange graph upstairs to the cluster exchange graph downstairs. In [50, Theorem 8.6], the authors show that the braid group acts fibrewise with respect to this map and moreover acts transitively on each fibre. This then establishes all the conditions needed for the argument of [12, Theorem 9.9]. □

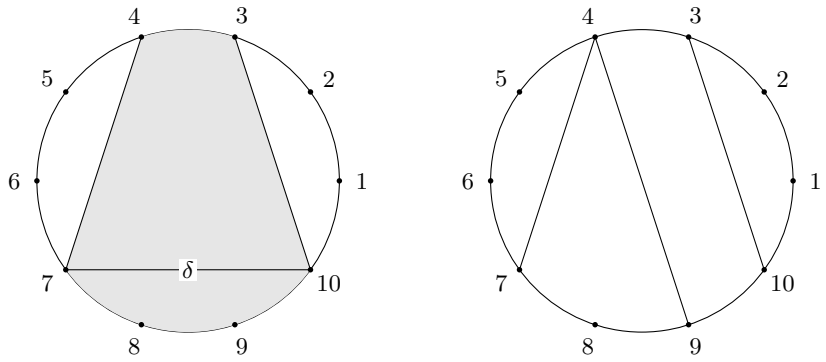


FIGURE 2. An n-angulation ($n = 4$, $k = 3$, $d_{k,n} = 10$) and its forward flip along the n-diagonal δ.

Lemma 2.8. *There is an equivalence* $\mathcal{R}\big(A_k^{(n)}\big) \simeq \mathcal{C}_{k,n}$.

Proof. This is observed *en passant* in the discussion before [54, Conjecture 1.5]. A more detailed identification of the cluster category and Rabinowitz Fukaya category for a Dynkin quiver is given in [7, Corollary 1.4]. □

Lemma 2.9. *There is a commuting diagram*

$$\begin{array}{ccccc} \mathrm{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big) & & \xrightarrow{\quad\text{Ikeda}\quad} & & \mathbb{Z}/d_{k,n}\mathbb{Z} \\ \uparrow & & & & \| \\ \mathrm{Auteq}\big(\mathcal{W}\big(A_k^{(n)}\big)\big) & \longrightarrow & \mathrm{Auteq}\big(\mathcal{R}\big(A_k^{(n)}\big)\big) & \longrightarrow & \mathbb{Z}/d_{k,n}\mathbb{Z}. \end{array}$$

Proof. On any Weinstein manifold, the Rabinowitz Fukaya category is algebraically determined from the wrapped category as the 'categorical formal punctured neighbourhood of infinity' (given by quotienting out proper modules), cf. [26] and the definition in Section 1, which defines the first arrow on the bottom line. The wrapped and compact Fukaya categories are in this case Koszul dual,[3] which defines the left vertical arrow.

Commutativity of the diagram amounts to the fact that both horizontal arrows are given by the action of autoequivalences on the heart exchange graph. In more detail, Ikeda [38, Section 7] constructs the top arrow directly using the action on the heart exchange graph. The quotient of the heart exchange graph by the braid group action is identified with the cluster exchange graph of $\mathcal{R}\big(A_k^{(n)}\big)$ by [50]. Furthermore, the cluster exchange graph is purely combinatorial, as discussed above, and the bottom right map is constructed using the action on the cluster exchange graph. □

Remark 2.10. It seems likely that the simplicial automorphism group of the cluster exchange graph is given by the mapping class group of the marked surface, which in this case is just the given cyclic group. This is known in the case of triangulations (here corresponding to the 3-fold case), but does not seem to have been written down for m-angulations in general. That would give a more direct argument for the previous result.

[3]Since $n \geqslant 3$, the compact category $\mathcal{F}(A_k^{(n)})$ is a Koszul dualising subcategory of $\mathcal{W}(A_k^{(n)})$; this follows by combining results of [2] and [42], cf. the discussion of Koszul duality later in Section 4 (which encompasses this case when $n = 3$, viewing the $A_k^{(n)}$-Milnor fibres as special cases of the 3-folds associated to marked bordered surfaces; the arguments when $n > 3$ are identical).

Since compactly supported symplectomorphisms of X act on both $\mathcal{F}$ and $\mathcal{W}$ and act trivially on the quotient category $\mathcal{R}(X)$, we immediately infer that the composite map

$$\pi_0 \operatorname{Symp}_{\mathrm{ct}}(X) \to \operatorname{Auteq}\big(\mathcal{F}\big(A_k^{(n)}\big)\big) \to \mathbb{Z}/d_{k,n}\mathbb{Z}$$

vanishes, hence the first arrow lifts to Br_{k+1}, as required.

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